9. Phase rule is an important tool used for the quantitative treatment of systems in equilibrium.

It enables us to predict the conditions that must be specified for a system to exhibit equilibrium.

Two or more different phases are present in equilibrium to form a “heterogenous system”. Such system is studied by phase rule.

J. Willard Gibbs enunciated the phase rule in 1876 on the basis of Thermodynamic principles

This rule predicts qualitatively the effect of temperature, pressure and concentration on a heterogenous equilibrium.

Phase (P)

Phase is a homogeneous system which is separated from other phases of the system by means of a boundary. It is a physically distinct and mechanically separable part of a system. The phase has to fulfill the following conditions:

1. It should be physically homogeneous. It must have the same composition and physical properties in all its parts.

2. It should be separated from other phases of the system at equilibrium by surfaces of contact.

3. The phases of a system are in dynamic equilibrium with one another through the exchange of chemical species.

Common examples of some phases

1. In a system of water and ice all the ice pieces are in one phase and water in another phase, so there are two phases.

2. Two liquids which are completely miscible with each other, e.g. glucose in water, sodium chloride in water, ethanol and water possess one phase.

3. Two liquids immiscible in all proportions., e.g., benzene and water and oil and water contain two phases.

Component (C)

It is an element or a compound present in a system. The concentration of the component can undergo independent variation. It does not stand for the total number of constituents of the system. The number of components of a system at equilibrium is defined as the minimum number of molecular species in terms of which the composition of each phase may be expressed quantitatively.

Examples:

1. In the water system, the phases that occur are ice, liquid water, and water vapour:

It is a one-component system, as each phase is in a different physical form and the composition of each of these phases can be expressed in terms of the single constituent represented by H2O.

Thus, the composition of phases present at equilibrium can be fixed by two variables. Hence it is a two-component system.

Degree of Freedom (F)

The degree of freedom or variance (F) is defined as the smallest number of independent variables such as pressure, temperature, and concentration that must be specified in order to describe the state of the system completely. The degrees of freedom of a system may be one, two, three or zero. Accordingly, the systems are known as uni-variant (or mono-variant), bi-variant, tri-variant and non-variant systems, respectively. For example:

1. Consider a gaseous system containing CO2 and nitrogen. The system is completely defined when three variables of temperature, pressure and concentration are specified. Then the system is said to have three degrees of freedom, i.e., F = 3.

Phase rule

Willard Gibbs in 1876 deduced the phase rule from thermodynamic considerations. This rule gives a relationship between the number of degrees of freedom (F), the number of components (C) and the number of phases (P) in a heterogeneous system at equilibrium. The relationship is as follows:

$$F = C - P + 2$$

One-Component System: The Water System

Under normal conditions the system 'water' is a three-phase, one-component system. In this system, water vapors, water, ice exist in equilibrium. All the three phases can be represented by one chemical formula H2O. Hence, it is a one-component system. If three phases exist together,

we have P = 3, C = 1 and the phase rule

$$F = C - P + 2$$

$$= 1 - 3 + 2$$

$$= 0$$

The system therefore has no degrees of freedom. It is called a non-variant or invariant system, i.e. none of the variables, i.e., pressure, temperature, concentration, can be varied without one of the phases disappearing from the system. Thus, the three phases can only exist together under one condition of temperature and pressure. If only two phases exist together, say ice and water or ice and vapour or water and vapour, the system now has one degree of freedom and is said to be univariant.

This means that one of the degrees of freedom, say temperature, may be varied without causing any alteration in the number of phases:

F = C – P + 2

C = 1, P = 2

F = 1 – 2 + 2 = 1

When pressure and temperature are required to define the system completely, the system represented by areas has two degrees of freedom and is bivariant:

F = C – P + 2

C = 1 P = 1

F = 1 – 1 + 2 = 2

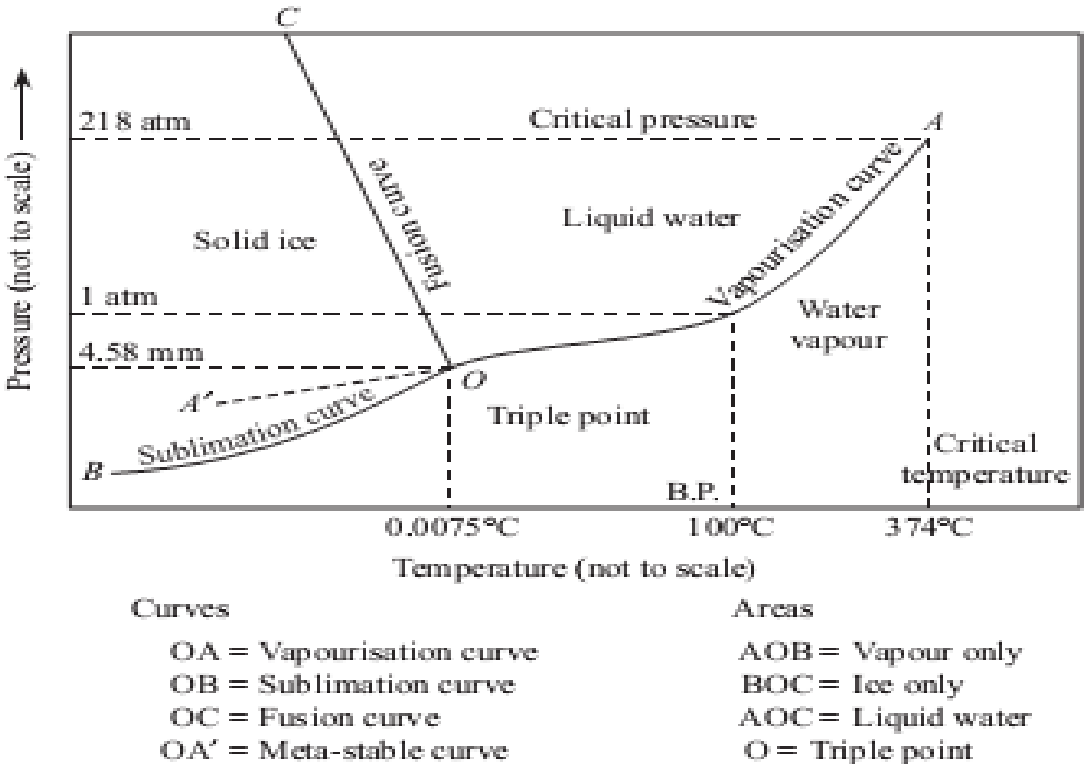


Fig. 7.1 Phase diagram for water system

Curves OA: This curve is known as vaporization curve as it represents equilibrium between liquid water and vapour at different temperatures:

Liquid water vapour the curve starts from '0', the freezing point of water and ends at A, the critical temperature of water, 374°C. Beyond point A, the two phases, liquid water and vapour, merge into each other, i.e. the number of phases it takes is two because liquid water and vapour exist along the curve OA. For any given temperature, there exists a fixed value of pressure and for each vapour pressure, temperature, has also a fixed value. Thus, the degree of freedom on any point of this curve is 1 or it is univariant, F = 1. Therefore,

F = C – P + 2

$= 1 – 2 + 2 = 1$

Curve OB: OB is the sublimation curve or the vapour pressure curve of ordinary ice. Along this curve, the two phases of solid ice are in equilibrium with its vapour. This curve shows vapour pressure of ice at various temperatures. The curve starts at O and ends at B, i.e. absolute zero (–273°C). At this phase, no vapour can exist and, therefore, only ice is present. On the other side of OB, ice is in equilibrium with vapour. Therefore, two phases exist and the degrees of freedom is 1 each.

$F = C – P + 2$

$= 1 – 2 + 2 = 1$

Vapour pressure can be maintained only at a fixed temperature.

Curve OC: OC is the melting point curve or fusion curve of ice. It indicates the effect of pressure on the melting point of ice. The fact that the curve OC is inclined towards the pressure axis indicates that the melting point of ice is lowered by an increase in pressure. At any point on curve OC, ice and liquid water exist in equilibrium. It has one degree of freedom and the system is univariant. The melting point must have a definite value.

Triple Point O- The three curves OA, OB, and OC meet at point O, the triple point. Here all the three phases, namely, ice, water and vapour co-exist. Further, P = 3, since there are three phases and one component. So

$F = C – P + 2$

$= 1 – 3 + 2 = 0$

In case of ice at the triple point, the system is non-variant. This occurs at a temperature of 0.0075°C and pressure 4.5 mm of Hg. Phase would not exist and one of the phases would disappear if temperature and pressure are varied from this value.

In the area AOB, only vapour exists and in the area AOC, only water exists. In the area BOC, three areas are bivariant and in order to locate any point in the area, temperature and pressure must be defined.

$F = C – P + 2$

$= 1 – 1 + 2 = 2$

Areas AOC, AOB, BOC

The areas between the curves show the conditions of temperature and pressure under a single phase–ice water or vapour is capable of stable existence.

Area AOC represents stable conditions for one-phase system, i.e. water. Area AOB represents conditions for the one-phase system water vapour. The area BOC represents the condition for one-phase system, i.e. ice. In all the three areas there being one phase and one component, we have

$$F = C - P + 2$$

$$= 1 - 1 + 2 = 2$$

Thus, each system, water, water vapour, or ice, has two degrees of freedom, i.e. the system is bivariant.

10. Two-Component System:

When a single phase is present in a two-component system, the number of degrees of freedom is F= 2 – 1 + 2 = 3, i.e., the three variables temperature, pressure and concentration of one of the components must be specified. Construction of the phase diagram of a two-component system will require besides temperature, pressure coordinates, the use of one concentration coordinate, which is possible in the space model or solid model.

A two-component system can be simplified by considering various types of equilibria separately. These are (1) liquid–gas (2) solid–gas (3) liquid–liquid and (4) solid–liquid equilibria. Solid–liquid systems are of great importance because they are connected with all crystallization problems.

These systems are characterized by the absence of a gas phase and are little affected by small changes in pressure. Systems where the gas phase is absent are called condensed systems. The measurement on solid–liquid equilibrium in condensed systems is carried out at atmospheric pressure. The phase rule takes the form F= C– P+ 1.

The reduced phase rule is more convenient to apply to a solid–liquid two-component condensed system. The variables are temperature and concentration of one of the constituents. Solid–liquid equilibrium is represented on temperature composition diagrams.

Simple eutectic system

Eutectic means easy to melt. A binary system in which two components react chemically and each component reduces the freezing point of the other is called eutectic system. In such a system, the two components are miscible in all proportions, i.e., eutectic system is a binary system consisting of two different, totally immiscible components in solid and miscible in liquid state without any chemical interaction.

The solid solution of a two-component system having lowest freezing point is called eutectic mixture. The lowest melting point that can be obtained corresponding to eutectic mixture is known as eutectic point.

The eutectic mixture possesses sharp melting point and definite composition. But it cannot be regarded as a com-pound, because (1) the compound does not possess components in stoichiometric proportion and (2) when the components are examined under a microscope, they appear as separate crystals of the compound.

Consider a two-component system having two components A and B. In liquid state, these components are completely miscible and these solutions on cooling yield pure A or pure Bas solid phases. Figure 7.3 consists of

Curve AC, the freezing point curve of A. On addition of Bto A, the freezing point of A falls along curve AC. Along this curve, solid Ais in equilibrium with the liquid solution of Bin A.

Curve BC, the freezing point curve of B. The curve exhibits decrease in freezing point by the addition of A to B. Along this curve, solid Bis in equilibrium with the liquid solution of Ain B.

When reduced phase rule equation is applied,

F= C– P+ 1 = 2 – 2 + 1 = 1

The degree of freedom is 1, i.e., both equilibria are monovariant.

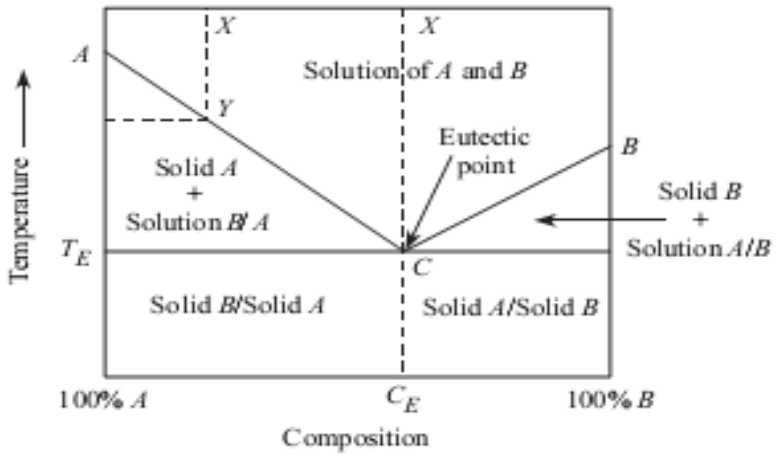


Fig. 7.3 Simple eutectic diagram

Eutectic point C: The curves AC and BC intersect at the eutectic point C, and the three phases—solid Ag, solid Pb and solution— are in equilibrium. Applying reduced phase rule to the system,

F= C– P+ 1 = 2 – 3 + 1 = 0,

i.e., At the eutectic point C, the system Ag/Pb/solution is non-variant.

Both the variables, temperature (303°C) and composition (97.4% Pb and 2.6% Ag), are fixed. If the temperature is raised above eutectic temperature, the solid phases Ag and Pb disappear and if the system is cooled below the eutectic point, only solid Ag/Pb exists where solution phase is non-existent. Area above AOC: This region is a single-phase system. The solution of molten Ag and Pb exists. When condensed/reduced phase rule is applied,

F= C– P+ 1 = 2 – 1 + 1 = 2.

Thus, Ag/Pb system is bivariant. The area below AC represents the phase Ag+ solution and that below BC, the phase Pb+ solution. The area below 303°C represents solid Ag+ and solid Pb. All these areas have two phases and one degree of freedom.

F= C– P+ 1 = 2 – 2 + 1 = 1

11. **Polymerization: -** It is defined as the process of linking or joining together smaller molecules i.e monomers to make large molecule.

There are mainly two types of polymerization reactions.

I. Chain polymerization (or) Addition Polymerization.

II. Step polymerization (or) Condensation polymerization.

I. Chain polymerization (or) Addition polymerization:

A chain Polymerization is a reaction that yields a polymer product which is exact multiple of monomers. Thus, the mechanism is also called addition polymerization.

Characteristics:

- The functionality of the monomer is two or bifunctional.
- The polymerization takes place by self-addition of the monomer molecules to each other.
- An initiator is required to start the polymerization reaction.
- The conversion of "=" to" –"takes place during the polymerization liberating 20 Kcal/mole of energy. Hence it is exothermic reaction.
- The molecular weight of the polymer is the exact multiple of the monomers.
- This mechanism is carried out in three steps. i.e.

  Initiation. ii)Propagation. iii)Termination.

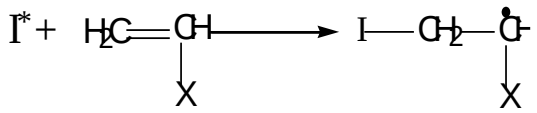
Chain polymerization is of three types

1. Free radical chain polymerization.
2. Ionic chain polymerization.
3. Co-ordination chain polymerization.

1) Free radical chain polymerization: The homolytic fission of a bond gives free radicals

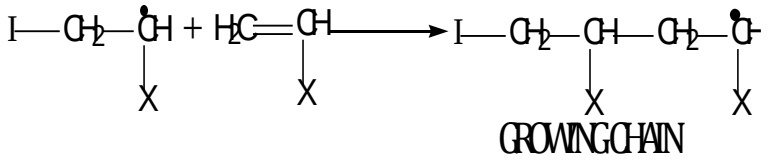
a.) Initiation: Initiators are unstable compounds and undergo homolytic fission to produce free radicals which reacts with π electrons of the monomer to produce monomer free radicals.

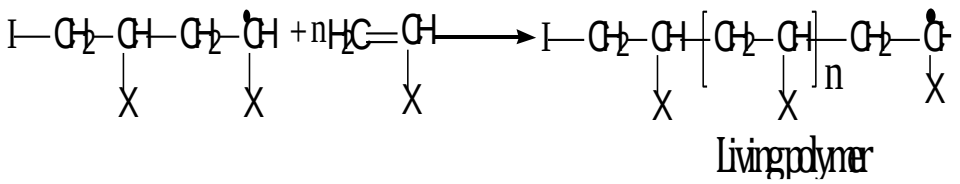
$$I_2 \xrightarrow{h\nu\Delta} I^{*} + I^{*}$$



Monomer free radical

b.) Propagation: The monomer free radicals react with a number of monomers rapidly resulting the chain growth with free radical site at the end of the chain producing a living polymer.



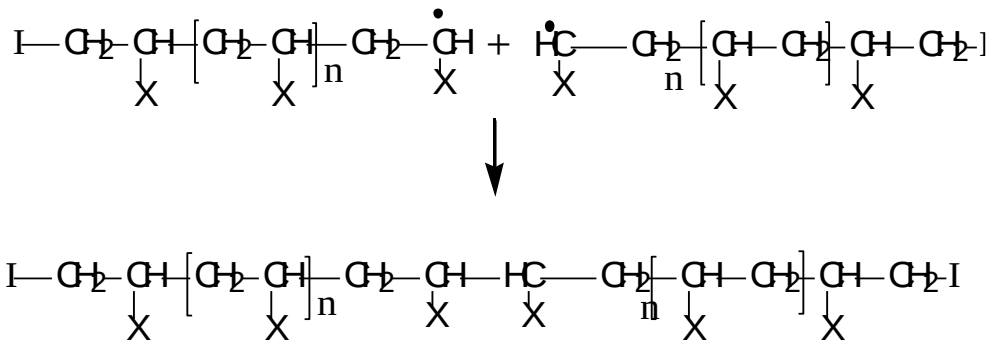


By adding fresh monomer to the living polymer with free radical site again chain growth starts. Hence it is known as living polymer

c) Termination:

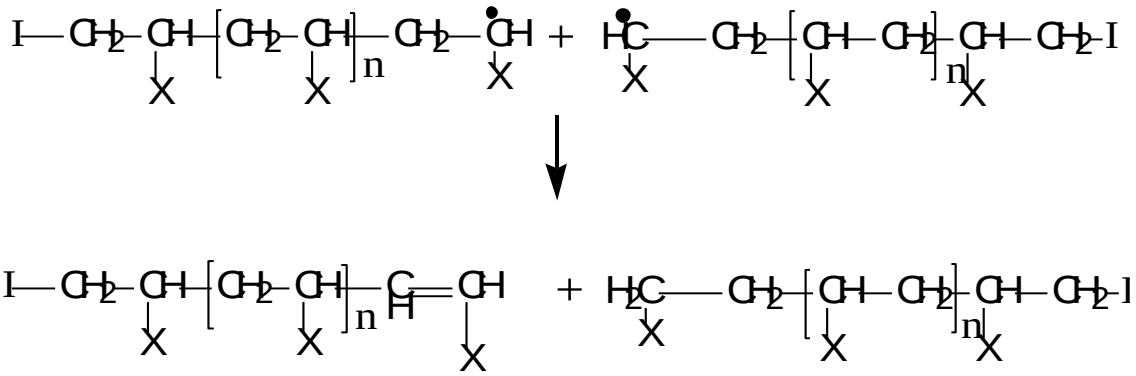
Termination takes place in two ways.

1).Self termination: In this step one growing polymer combine with other growing polymer produce a dead polymer.



Dead polymer

2) Termination by disproportionation: Hydrogen from the growing chain is abstracted by the other growing chain, utilizing the lone electron for getting stabilized.



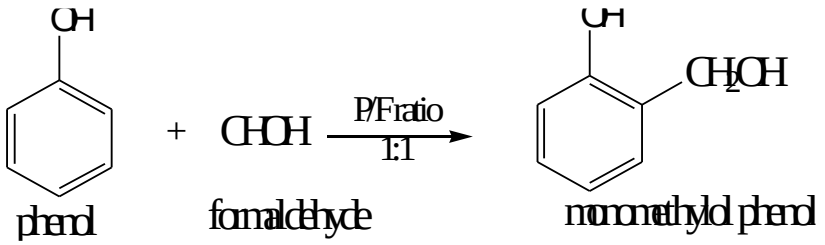
12. **Bakelite or (Phenol- Formaldehyde Resin):**

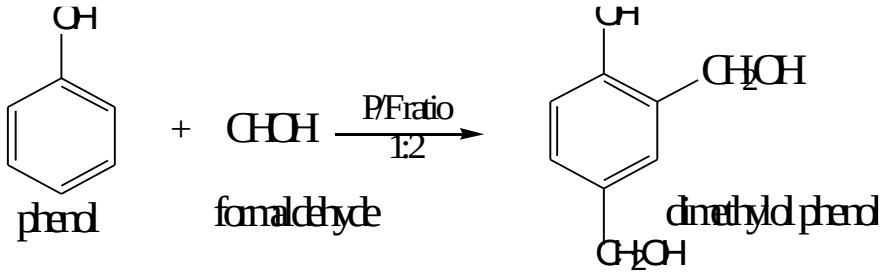
Bakelite is an important Thermoset resin. It is prepared by the step polymerisation of phenol with formaldehyde in presence of an acid or alkali as a catalyst.

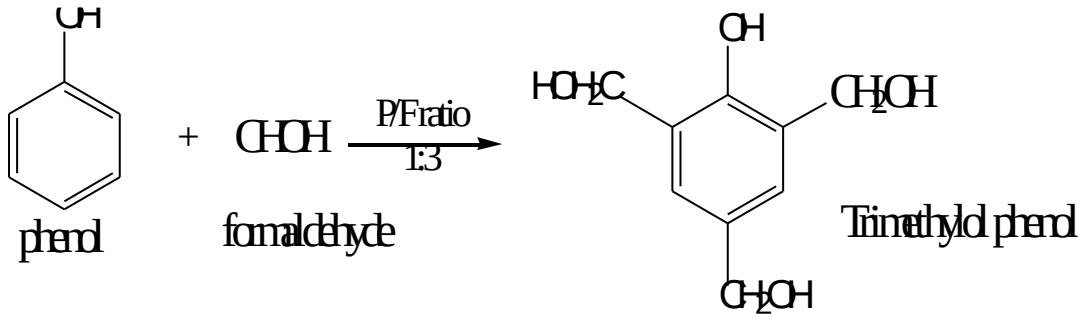
The polymerization takes place in three stages:

Stage-i:

Phenol is made to react with formaldehyde in presence of acid/alkali to produce mono, di, tri methylol phenols depending on the phenol formaldehyde ratio (P/F ratio)

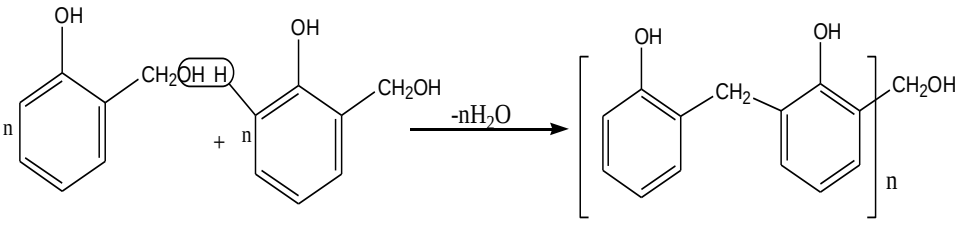




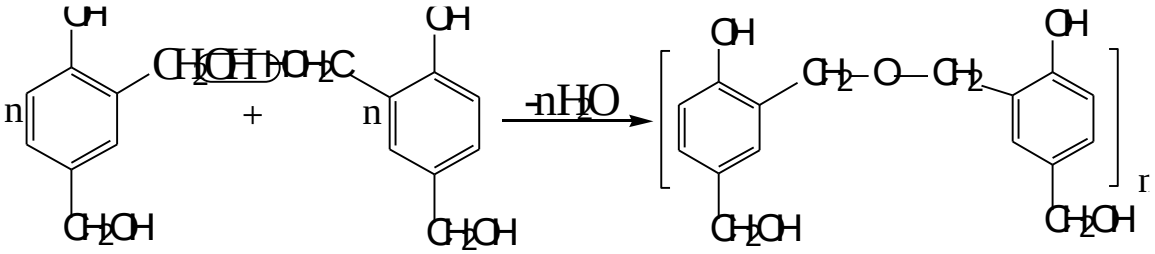


Stage-ii:

The mono, di, and tri methylol phenols are heated to produce two types of straight chain resins by condensation of the methyl group with hydrogen atom of benzene ring or another methyl group.



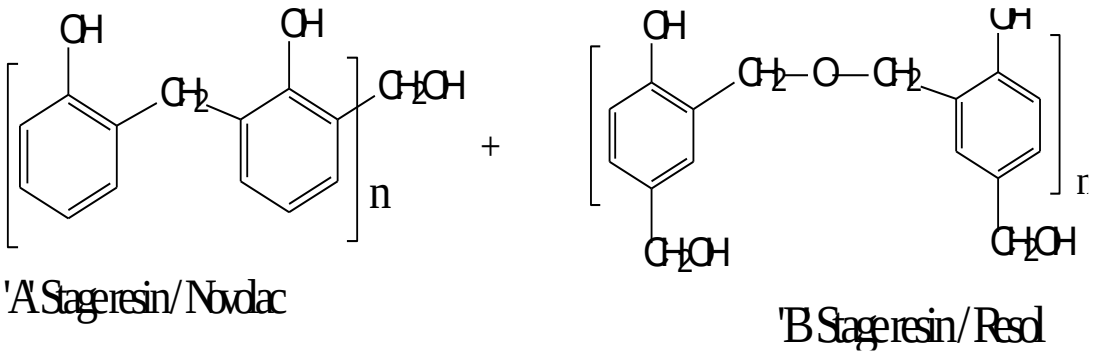
'A' Stage resin / Novolac

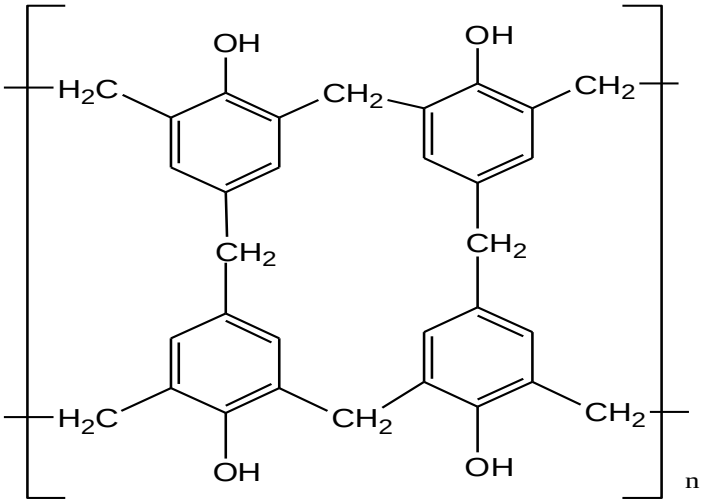


'B' Stage resin / Resol

Stage iii:

This stage of preparation includes heating of ‘A’ stage resin and ‘B’ stage resin together which develops cross linking and produce Bakelite plastic resin.





Bakelite

Properties: -

- Bakelite is hard, rigid and strong, it is a scratch resistant and water-resistant polymer.
- It is resistant to chemicals but it is attacked by alkalis due to presence of OH group.
- Bakelite is an excellent electrical insulator and has very good corrosion resistance.
- It is a good anion exchange resin, exchanges –OH group with any other anion.

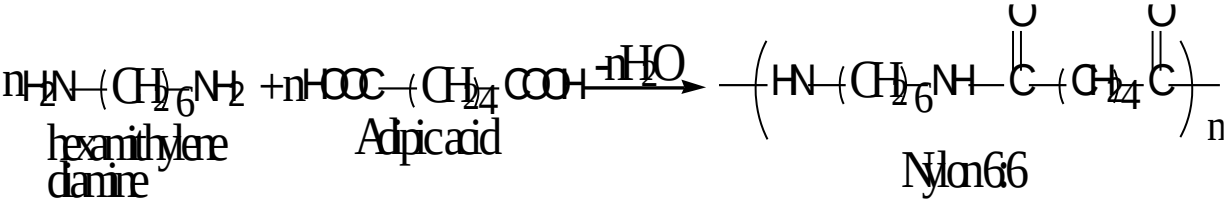
Applications:

- It is used for making electrical insulator parts like switches, heater handles.
- It is used for molded articles like telephone parts, cabinets for radio and television.
- It is used for making bearings used in propeller shafts, paper industry and rolling mills.

- It is used in paints and varnishes

Nylon 6:6

Nylon 6:6 is produced by the co-polymerization of hexamethylene diamine with adipic acid.



Properties:

- Nylon possesses high strength, hardness and high melting point. The high melting point of the nylon is due to the hydrogen bonding between the hydrogen atom of the amide group and oxygen of c = o groups.
- Nylon possesses good chemical resistance and electrical resistance.
- Nylon possesses good resistance to environmental conditions.
- They are good insulators.

Applications:

- The major application is in the textile industry.
- Because of its high thermal and abrasion resistance nylons are used in mechanical engineering applications like gears, bearings etc.
- Flexible tubing's for conveying petrol etc. are made from nylons.
- Nylons are used as electrical insulators.
- Nylon 6 is used for making tire cords.
- Nylons are used in automobile industry and telecommunication industry for making radiator parts and coil formers respectively.

13. **Conducting polymers**

The polymers which conduct electricity is called conducting polymers. The conduction of the polymers may be due to unsaturation or due to the presence of externally added ingredients in them.

The conducting polymers can be classified in the following way.

Conducting polymers

Intrinsic conducting polymers

Intrinsic polymer conducting with conjugation polymer

Doped conducting Polymer

Extrinsic conducting polymers

Conductivity Element

Blended

Doped conducting polymers: -

The conductance of a polymer can be increased by introducing a positive charge or negative charge on polymer backbone by oxidation reduction. This process is similar to semiconductor technology and is called doping.

Doping is two types: -

1. Creating a positive site on polymer backbone called p-doping.
2. Creating a negative site on the polymer backbone called n-doping.

<u>Doping and conductivity of polyacetylene</u>

P-doping: -

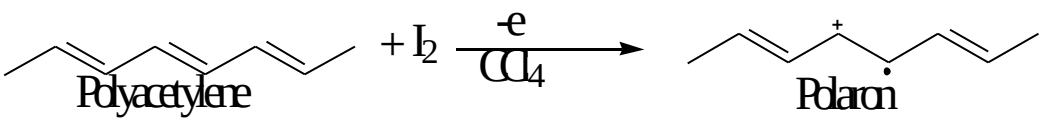
P-doping is done by oxidation of a conducting polymer like polyacetylene with a Lewis acid or iodine vapors.

This is also called oxidative doping.

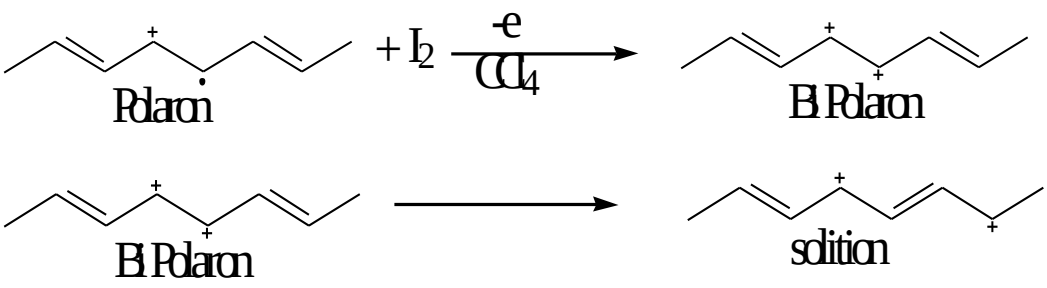
1. $$-[HC{=}CH]_n- + 2FeCl_3 \longrightarrow -[HC{=}CH]_n^{+}\ FeCl_4^{-} + FeCl_2$$

2. $$-[HC{=}CH]_n- + 3I_2 \longrightarrow -[HC{=}CH]_n^{+}\ I_3^{-}$$

In the above reaction the Iodine molecules attract an electron from polyacetylene chain and become I_3^{-} now the polyacetylene becomes positively charged and termed as radical cat ion or polaron.

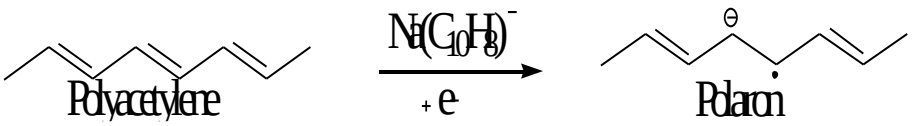


The polarons on doping forms solitons i.e. . Further oxidations of polaron give solitons.

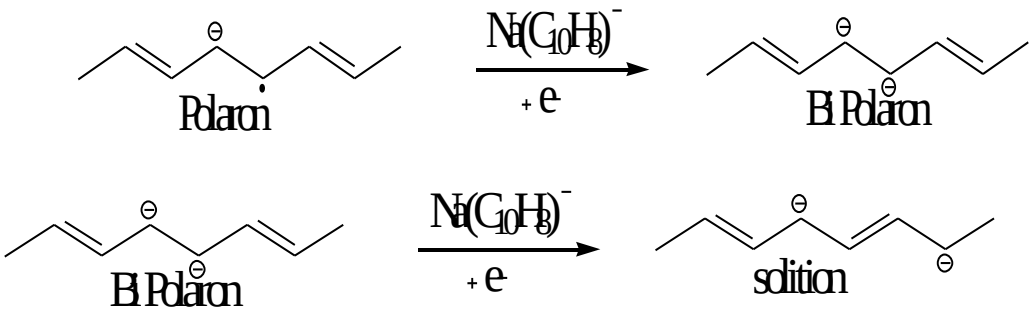


N-doping:

N-doping is carried out by reduction process by the addition of an electron to polymer backbone by using reducing agents like sodium naphtalide $Na^{+}(C_{10}H_{8})$. Formation of polaron, bipolaron takes place in two steps, followed by recombination of radical which yields two charge carriers on the polyacetylene chain responsible for conduction as shown below



The electron added to polyacetylene by reductive doping doesn't go into the conductivity band but goes into an intermediate electronic state within the band gap of radical anion (polaron).



The bi anion lowers the energy by segregating into two negative solitons at the midgap energy levels. This leads to the generation of conduction pathways. As a consequence, the conductivity increases significantly.

14. **Analysis of Coal :**

The analysis of coal is helpful in its ranking. The assessment of the quality of coal is carried out by these two types of analysis

i) Proximate analysis

ii) Ultimate analysis

(i) proximate analysis: In this analysis, the %c is indirectly determined.

This analysis includes %moisture, volatile substance, ash content and carbon.

i. Moisture:

A known mass of finely powered coal is taken in a crucible. It is heated up to 110^0C for an hour and cooled to room temp. in a desiccator. The moisture is removed as water vapors and the process is repeated till the consent wt. is obtained.

$$\% \text{ Of moisture} = \frac{loss\ in\ weight\ of\ coal}{weight\ of\ coal} \times 100$$

ii. Volatile matter:

The above sample is taken and heated at 950^0 C in the absence of air for 7min. It is then cooled to room temp. and weighted. The loss of wt. is noted as volatile matter and is removed from coal.

$$\% \text{ Of volatile matter } = \frac{loss\ in\ weight\ due\ to\ removal\ of\ volatile\ matter}{weight\ of\ coal} \times 100$$

iii. Ash Content:

Coal free from moisture and volatile matter is heated in a crucible at about 700^0C in presence of air.It undergoes combustion and results formation of ash. Crucible is cooled room temp. and weighted .The mass of ash is then determined.

$$\% \text{ of ash} = \frac{mass\ of\ ash}{mass\ of\ coal} \times 100$$

iv. Carbon:

Since main component of coal is carbon, it can be determined by subtracting the sum of the % of moisture, volatile substance and ash content from 100.

$$\% \text{ C } = 100 - (\% \text{ of moisture} + \% \text{ of volatile matter} + \% \text{ of ash})$$

Significance Of Analysis:

Moisture:

high moisture content in the fuel reduces the calorific value, increases the cost of transportation and causes wastage of heat. Hence lesser the moisture content the better is the quality of fuel.

Volatile Matter:

A coal containing high volatile matter burns with long flame, high smoke and low calorific value .Volatile matter also influence the design of the furnace since the higher the volatile matter, larger is the combustion space required.

Ash:

Ash reduces heating value of coal. Ash content increases cost of transportation, handling, storage and disposal. it determines the quality of coal. Hence lesser % of ash, the better is the quality of coal.

Fixed Carbon:

Higher the fixed carbon in a coal, greater is its calorific value.

II.ULTIMATE ANALYSIS:

This analysis includes % of C, H, O, S, N and ash content in coal.

1. C and H determination:

A known mass of carbon is taken and burnt in an apparatus. The C change in to CO_2 and H changes in to HO_2.The vapors of CO_2 and H_2O are then passed through KOH and $CaCl_2$.the CO_2 is absorbed by KOH and H_2O is absorbed by $CaCl_2$ There by weight of KOH and $CaCl_2$ increases which is then measured.

$$C + O_2 \longrightarrow CO_2$$

$$H_2 + 1/2\ O_2 \longrightarrow H_2O$$

$$2KOH + CO_2 \longrightarrow K_2CO_3 + H_2O$$

$$CaCl_2 + 7H_2O \longrightarrow CaCl_2\ 7H_2O$$

$$\%\ \text{of C} = \frac{increase\ in\ wt\ of\ KOH}{wt\ of\ Coal} \times \frac{12}{44} \times 100$$

$$\%\ \text{of H} = \frac{increase\ in\ wt\ of\ cacl2}{wt\ of\ coal} \times \frac{2}{18} \times 100$$

Significance:

Higher the % of C and H, the better is the quality of coal and higher is the calorific value. % C helps in assessing the ranking of the coal.

2. Nitrogen:

The estimation of N is done by kjeldahls method. About 1g of finely powdered coal sample is heated with conc. H_2SO_4 along with K_2SO_4 catalyst in a kjeldahls flask. N in the coal is converted in to $(NH_4)_2SO_4$.

$$2N+3H_2+H_2SO_4 \longrightarrow (NH_4)_2SO_4$$

$(NH_4)_2SO_4$ is heated with excess of NaOH and the liberated NH_3 is distilled over and absorbed by a known vol. of std. acid sol. The untreated acid is back titrated with std. NaOH. The amount of acid neutralized by liberated ammonia is determined.

$$(NH_4)_2SO_4 + 2NaOH \longrightarrow 2NH_3+Na_2SO_4+2H_2O$$

$$NH_3+ HCl \longrightarrow NH_4Cl$$

From this the % of N is calculated.

$$\% \text{ of N} = \frac{volume\ of\ H2SO4 \times normality}{wt\ of\ coal} \times 1.4$$

Significance:

N does not have any calorific value. It has no significance.

<u>3. Sulphur</u>:

It is estimated gravimetrically in terms of $BaSO_4$.sulphur is converted into soluble sulphate by incinerating coal with 1:2 mixture of Na_2CO_3 and $MgCO_3$. Then treated with $BaCl_2$ sol. So that sulphates are precipitated as $BaSO_4$ .From the wt. of $BaSO_4$ obtained, the S present coal in is calculated.

$$\% \text{ of S} = \frac{wt\ of\ BaSO4}{wt\ of\ coal} \times \frac{32}{233} \times 100$$

Significance:

- S increases calorific value
- The product of combustion SO_2 , SO_3 have corrosive effect on equipment and cause air pollution.
- S is undesirable in the preparation metallurgical coke.

<u>4. Ash Content</u>:

The coal sample is ignited and the wt. of ash is measured at room temp.

$$\% \text{ of ash} = \frac{wt\ of\ ash}{wt\ of\ coal} \times 100$$

5. Oxygen:

The % of O_2 is determined by subtracting the sum of % of C, H , S and ash from 100.

$$\% \text{ of } O_2 = 100 - (\% \text{ of C} + \%H + \%N + \%S + \%A)$$

Significance:

High O_2coal contains high inherent moisture and low calorific value. A quality coal should contain low % of O_2.

15. Refining of crude petroleum:

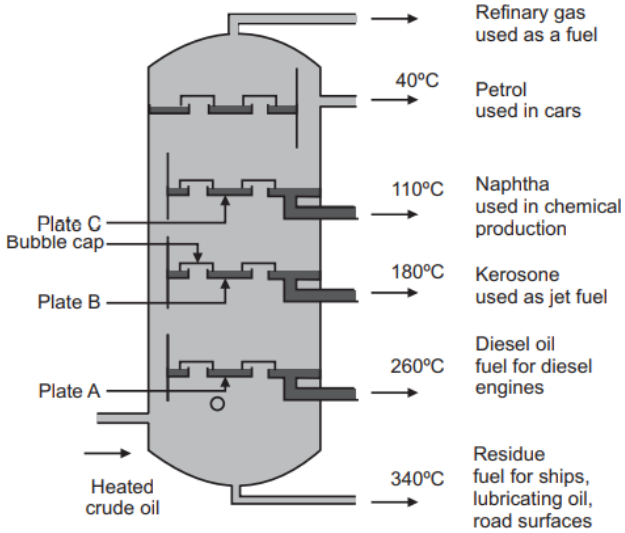
The crude oil is a mixture of several chemicals. They are separated by three steps. This process of separating various fractions of crude oil is called refining and industry is called oil refineries.

1) **Separation of water (Cottrell's process):**
   The crude oil is an emulsion of oil and water. Water is separated by allowing the crude oil to flow through two highly charged electrodes. The colloidal water is separated from oil in the form of big droplets.

2) **Removal of Sulphur compounds:**

The crude oil is treated with copper oxide which gives black CuS precipitation can be removed by filtration.

3) **Fractional distillation:**
   Crude oil obtained after 1 and 2 steps then heated at about 400°C in an iron retort. All volatile constituents are evaporated leaving behind some residence (asphalt or petroleum coke). The hot vapors are passed through different fractionating columns. The apparatus is a long cylindrical vessel with several trays, each having chimney covered with a loose cap.



16. Synthetic Petrol (gasoline):

Petrol is mainly obtained from crude oil from oil wells but can also be obtained from synthetic process. There are two types of synthetic processes. They are,

1. Fisher-Tropsch process
2. Bergius process

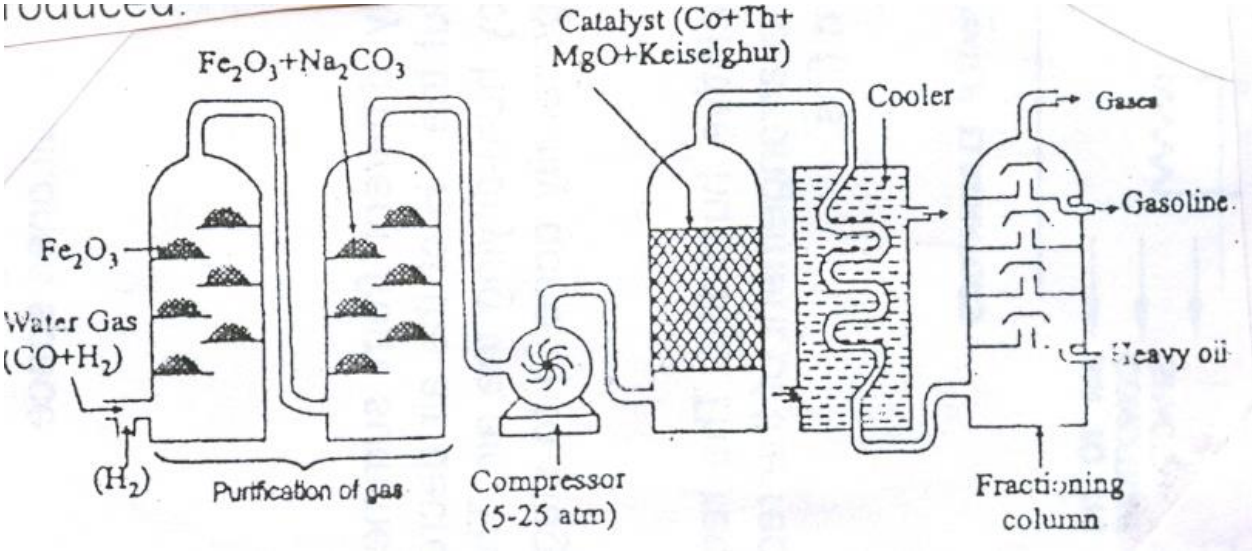
1. Fisher-Tropsch process:

This process was developed in Germany by F. Fisher and H. Tropsch in 1923. In this process coal is converted to coke. Coke is then treated with steam to obtain water gas.

$$nCO + (2n+1)\,H_2 \longrightarrow C_nH_{2n+2} + nH_2O$$

$$nCO + 2nH_2 \longrightarrow C_{2}H_{2n} + nH_2O$$

Water gas produced by passing steam over heated coke is treated with hydrogen in the presence of group VIII metals (Fe, Co, Ni) which give gasoline and fuel oil.



In order to remove H_2S and organic S compounds, the gas is passed through Fe_2O_3 and then into a mixture of $Fe_2O_3+Na_2CO_3$. Pure gas is then compressed at 5-25 atm and led to a convertor (packed with mixture of 100 parts of Co, 5 parts of Ni, 8 parts of MgO and 200 parts of Keiselguhr earth. The hot gaseous mixture is then cooled and fractionated and the heavy oil fraction is cracked to get gasoline.

These reactions take place at 30 atm pressure at 200-300° C temperature. The gasoline formed by this process has low anti-knock value.